

INDIAN STATISTICAL INSTITUTE
End-Semester Examination
Bachelor of Statistics and Data Science 2nd Year
Data Structures and Algorithms

Date: 12.12.2025
Duration: 3 Hours

Total Marks: 100

Note: Answer all questions. The maximum you can score is 100.

1. (a) Insert the following keys into an empty Binary Search Tree in the given order:

50, 30, 70, 20, 40, 60, 80, 65

Draw the BST after each insertion.

- (b) Convert the above BST into an AVL tree by performing necessary rotations. Show every imbalance and rotation clearly.
- (c) Prove that the height of an AVL tree with n nodes is $O(\log n)$.

[6+9+5=20]

2. A hash table of size $m = 17$ is used with hash function $h(k) = k \bmod 17$. The following 15 keys are inserted in the given order:

42, 23, 34, 52, 71, 15, 88, 60, 27, 94, 39, 10, 56, 75, 91

- (a) Construct the hash table using separate chaining.
- (b) Construct the hash table using linear probing. Show all collision resolution steps.
- (c) Explain deletion in linear probing with a proper example.

[7+8+5=20]

3. Let $G = (V, E)$ be an undirected graph with $|V| = n$ and $|E| = m$.

- (a) Write the algorithm for DFS of an undirected graph.
- (b) Using DFS, design an algorithm to find all connected components of an undirected graph G .
- (c) Let $G = (V, E)$ be a directed graph, and let $T = (V, F)$ be a DFS tree of G . Prove that if (u, w) is an edge of E such that $u.DFS_{Number} < w.DFS_{Number}$, then w is a descendant of u in tree T . (For $u \in V$, $u.DFS_{Number}$ denotes the DFS number of the node u).

[5+7+8=20]

4. Consider the weighted directed graph with source s :

$$V = \{s, a, b, c, d, e\}$$

$$E = \{(s, a, 4), (s, b, 1), (b, a, 2), (a, c, 3), (b, c, 6), (c, d, 2), (a, d, 10), (d, e, 3)\}$$

- (a) Write the algorithm for Dijkstra's single source shortest path algorithm.
- (b) Apply the algorithm step-by-step on the above graph using a priority queue. Show all intermediate distance updates.
- (c) Construct the shortest path tree.

[8+8+4=20]

5. Let $G = (V, E)$ be a connected undirected weighted graph.

- (a) Write Kruskal's algorithm using the Union-Find data structure.
- (b) Apply Kruskal's algorithm on the following edge set and find the MST:

$$(a, b, 2), (a, c, 3), (b, c, 1), (b, d, 4), (c, d, 5), (c, e, 6), (d, e, 7)$$

Show all Union and Find operations.

- (c) Apply Prim's algorithm on the same graph starting from vertex a .

[5+7+8=20]

6. (a) Compute the 4-point FFT of the vector $(0, 2, 4, 7)$. Show all intermediate stages.
- (b) Construct the pre-computation for the pattern $P = A B A B A C A B$ and apply the KMP algorithm to find its occurrences in $T = ABA B A B A C A B$. Show all shifts.

[8+12=20]